

2024



Nykaa Builds Unified Data Platform for Insights Generation Using AWS

Find out how the Indian omnichannel retailer of beauty, fashion, and wellness brands scaled using AWS to generate business insights and achieve strategic growth through increased scalability and cost savings.

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Unified Insights

Single source of truth allows for tailored



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30%

Reduction in core software expenses for concept



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



Faster



Report configuration from 2-3 weeks on the previous framework down to 4-5 hours

48x

Increase in daily data refreshes, which facilitates more accurate and timely reporting

Overview

In 2012, [Nykaa](#) embarked on a journey to reshape India's retail landscape as one of the country's first beauty e-commerce platforms. Over the last decade, Nykaa expanded its product offerings by introducing two platforms—Nykaa Fashion and Superstore by Nykaa. Today, through its websites, mobile apps, and a network of over 165 physical stores countrywide, Nykaa retails diverse products encompassing beauty, fashion, and wellness. With an extensive repertoire today of over 6000 brands, Nykaa serves over 40 million active users monthly.

Recognizing the need for robust data aggregation and analysis, Nykaa began building a data lake with common ingestion, governance, and query layers using Amazon Web Services (AWS). With AWS, Nykaa can now execute and scale its big data processing frameworks, thus aligning with the company's vision of “One Nykaa, One Data.” Since building the data lake, Nykaa has established a unified data platform while concurrently reducing report configuration times from weeks to just 4-5 hours.



Opportunity | Achieving Unified Data Insights Cost-Effectively with AWS

Over the past decade, as Nykaa evolved from an exclusively beauty and personal care e-commerce platform to a multi-vertical retail giant, it began to encompass various verticals, including grooming for men, fashion, eB2B and expanding its focus on wellness, digital marketing, and more. Finally,



data challenges arose. Each vertical operated with distinctly separate databases, which prevented the company from accessing a single source of sales, customer, and product inventory data.

Nykaa's siloed systems impeded the company's ability to perform effective data analytics and generate business insights on its customers. Additionally, Nykaa's multiple, disparate databases resulted in time-consuming management and increased the risk of contradictory data. The company also found that duplicating the same data across various locations increased its overall storage footprint and led to inefficient resource utilization.

Furthermore, for performance monitoring, Nykaa regularly had to collate and analyze clickstream and transactional data across various verticals. This resulted in data fragmentation across different databases. The company needed a team of 2-4 staff up to 3-4 weeks to generate the relevant business reports. Due to the time-consuming process, Nykaa would sometimes miss out on opportunities to reach specific customer groups at specific times.

In 2022, Nykaa embarked on its "One Nykaa, One Data" initiative to create a unified data ingestion, governance, and query framework. The company turned to AWS to streamline its operations into a single, scalable, reliable, and cost-effective data lake. The goal was to consolidate data from diverse sources, to enhance transparency for all stakeholders, yield superior business insights, and generate cross-selling and upselling opportunities.

Solution | Harnessing Deeper Insights for Agile Market Responses

In September 2022, Nykaa built a proof of concept (POC) by comparing the cost efficiency of [Amazon EMR](#)—a cloud-based big data solution for processing, analytics, and machine learning using open-source frameworks—against other cloud-based analytics solutions. After 3-4 weeks, Nykaa observed a 30 percent reduction in its core software expenses due to Amazon EMR's elastic scaling capabilities and pay-as-you-go pricing model.

In Mar 2023, Nykaa began using Amazon EMR, [Amazon Simple Storage Solution](#) (Amazon S3), an object storage service, and cloud data warehousing service [Amazon RedShift](#) to build and deploy the ingestion and storage layer of its data lake. As a cornerstone of its "One Nykaa, One Data" initiative, the data lake can now consolidate 100 TB of unarchived data from various sources into a single data lake.



With autoscaling built into Amazon EMR, the company can now automatically scale its data processing workloads, facilitating up to 10,000 data ingestion jobs. Consequently, Nykaa's business analytics team requires only 5 hours to generate a new business report from what took up to 2 weeks. Today, the team has scaled the data processing platform to generate over 1,500 sets of customer data. It also collects data on an hourly basis instead of daily, allowing Nykaa to swiftly respond to market trends and emerging customer preferences.

Nykaa uses [AWS Lake Formation](#) to build, manage, and secure the new data lake at the governance layer. With a secure way to gain deeper insights into customer behavior and preferences, Nykaa can build personalized marketing strategies and tailor product recommendations for different customer segments.

Finally, Nykaa uses Amazon Redshift and [Amazon Athena](#), a serverless analytics service, to analyze petabyte-scale data at the query layer. Since both services are fully managed, Nykaa's developers do not have to spend time operating and maintaining the data warehousing and query services.

Outcome | Navigating a Data-Driven Future with Enhanced Insights

The company aims to incorporate more data sources in future, including real-time feeds, to generate even more targeted insights and create personalized customer experiences.

With the consolidated data lake, Nykaa's vision extends to integrating machine learning capabilities. With this, the company can bolster its search and recommendation functionalities, optimize inventory management, and improve backend operations—all to elevate customer experiences.

"In India's increasingly crowded e-commerce market, the tiniest competitive edge matters significantly. Building on AWS, we now have a highly scalable and cost-effective data lake that is secure and easy to govern. This has helped us unlock profound customer insights that were previously beyond our reach," says Rajat Kumar, Head of Data Platform at Nykaa.

Learn More

To learn more, visit aws.amazon.com/retail.

About Nykaa

Nykaa (FSN E-Commerce) was founded in 2012 by Indian entrepreneur Falguni Nayar with a vision of bringing inspiration and joy to people, everywhere, every single day. Since then, Nykaa has emerged as one of India's leading lifestyle-focused consumer technology platforms and has expanded its product categories by introducing online platforms Nykaa Fashion, Nykaa Man, and Superstore. Delivering a comprehensive omnichannel e-commerce experience, Nykaa offers over 6000 brands on its website and mobile applications. The Nykaa Guarantee ensures that products available at Nykaa are 100% authentic and sourced directly from the brand or authorized retailers. Through engaging and educational content, digital marketing, social media influence, robust content, and the Nykaa Network community platform, Nykaa has built a loyal community of millions of beauty enthusiasts.



AWS Services Used

Amazon Simple Storage Solution

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

Amazon RedShift

Amazon Redshift uses SQL to analyze structured and semi-structured data across data warehouses, operational databases, and data lakes, using AWS-designed hardware and machine learning to deliver the best price performance at any scale.

[Learn more »](#)

Amazon EMR

Amazon EMR is the industry-leading cloud big data solution for petabyte-scale data processing, interactive analytics, and machine learning using open-source frameworks such as Apache Spark, Apache Hive, and Presto.

[Learn more »](#)

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